A Quantitative Convergence Law for the Connes–Consani–Moscovici Zeta Spectral Triple

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Abstract

The Connes–Consani–Moscovici (CCM) zeta spectral triple [1] and its Connes–van Suijlekom refinement [2] produce finite real symmetric matrices whose ground state yields an approximation ν_1 to the first nontrivial Riemann zero γ_1 that sharpens as three parameters grow: a prime cutoff λ^2, a basis size N, and the working precision P. The rate of this convergence in the basis size is named the central open problem in [1]. We characterize the convergence quantitatively, in two equivalent forms. Intrinsically, the construction's relative error $|\nu_1 - \gamma_1|/|\gamma_1|$ is a single two-channel quantity,

$$E(N, \lambda^2) = \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

a mode channel vanishing as $N \to \infty$ and a prime channel vanishing as $\lambda^2 \to \infty$, so that $\nu_1 \to \gamma_1$ is exactly $E \to 0$ and forces both limits. The two channels are the finite-N transition and the extreme tail of a single spectral edge (the lower edge of the equilibrium measure of the archimedean field, whose bulk density is the Riemann–von Mangoldt law), so D_{nModes} and D_{Primes} are one edge read at two scales. Observed at finite working precision P, the same law is the matching-digit count $D(\lambda^2, N, P) = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$,

$$D(\lambda^2, N, P) = \min\bigl(D_{\text{Wprec}}(P),\, D_{\text{nModes}}(N, \lambda^2),\, D_{\text{Primes}}(\lambda^2)\bigr),$$

the minimum of three budgets — precision, modes, and prime content — of which precision is the instrument resolution, absent from the intrinsic form. In digit space the minimum is forced, not fitted (D is a negative logarithm of a sum of independent errors, so the largest binds), which is why earlier single-expression digit fits fail. We prove that convergence to γ_1 requires all three budgets to diverge: at a fixed cutoff the accuracy is capped at the prime ceiling $D_{\text{Primes}}(\lambda^2)$, so no basis size or precision reaches the zero and the cutoff must grow. This reduces convergence to the growth of the prime ceiling, whose leading coefficient is derived, parameter-free, from the prolate spheroidal eigenvalue asymptotic: $D_{\text{Primes}} \sim (4\pi/\ln 10)\lambda^2$. The mode budget — the basis-size convergence rate — is the logarithm of a stretched-exponential convergence $\exp(-aN^q)$ whose exponent $q(\lambda^2)$ is the analyticity class of the eigenvector, derived from the archimedean Gamma factor with a prime correction from Mertens' theorem; a by-product is that the eigenvalue-space Galerkin exponent measured by Groskin [7] is a finite-window slice of the same curve. We state each component with an explicit rigor level, separating what is proved versus what is derived and empirically validated up to $\lambda^2 = 200$. The results have been communicated to co-authors of [1].

1 Introduction

The Connes–Consani–Moscovici (CCM) construction [1], building on the Connes–van Suijlekom truncated Weil form [2], produces a family of finite self-adjoint operators $D_{\log}^{(\lambda,N)}$ whose spectra

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approximate the imaginary parts γ_k of the nontrivial Riemann zeta zeros to remarkable numerical accuracy — 55 digits at the headline configuration of [1], which we extended to 999 digits in an independent reproduction [16]. The construction has two integer-like control parameters — a prime cutoff λ^2 (the primes $p \leq \lambda^2$ enter the Weil form) and a basis size N (the matrix is $(2N{+}1) \times (2N{+}1)$) — together with the working precision P in decimal digits. The rate at which the eigenvalue converges in the basis size N is named the central open problem in [1]; Groskin [7] measured a Galerkin convergence exponent empirically but disclaims deriving it, and Śliwiński [8] proves an inverse-logarithmic lower bound on the mean spectral error. We previously tested two proposed rate laws against high-precision data and found them unsupported [17].

This paper characterizes that convergence explicitly.

Definition 1. $D(\lambda^2, N, P) := -\log_{10}\big(|\nu_1 - \gamma_1|/|\gamma_1|\big)$ *is the* matching-digit count*: the number of correct leading decimal digits of the first Riemann zero γ_1 reproduced by ν_1, the first ordinate produced by the CCM construction, computed at working precision P via MPFR/GMP. The reference γ_1 is computed to* 2500 *significant digits by rigorous interval arithmetic (Arb [18]), far exceeding the largest match reported here ($\approx$* 1070 *digits at $\lambda^2 = 200$), so the reported counts are genuine and not reference-limited; its leading digits agree with the standard tabulations of [19].*

Our central result, the *Andrews CCM Convergence Law*, has two equivalent forms. Its intrinsic (spectral) form is the two-channel error

$$\boxed{E(N, \lambda^2) = \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2)} \tag{1}$$

whose two channels are the finite-N transition and the extreme tail of a single prolate-type spectral edge (Section 3.1); convergence $\nu_1 \to \gamma_1$ is exactly $E \to 0$. Its finite-precision measurement is the matching-digit count

$$\boxed{D(\lambda^2, N, P) = \min\big(D_{\text{Wprec}}(P),\ D_{\text{nModes}}(N, \lambda^2),\ D_{\text{Primes}}(\lambda^2)\big)} \tag{2}$$

in which D_{Wprec} is the instrument resolution alone. Since D is the relative error on a logarithmic scale, the law is a quantitative convergence statement: $D \to \infty$ is precisely $\nu_1 \to \gamma_1$, and the three budgets are the independent mechanisms that drive (or stall) that convergence (Prop. 2). Five points distinguish it from prior empirical fits:

1. **The minimum is forced, not fitted** (Section 3). In digit space D is a negative logarithm of a sum of independent positive errors, so the largest dominates and D equals the minimum of the three per-source budgets up to an $O(1)$ term — the structural reason earlier single-formula digit fits failed. The precision budget is measurement, not mathematics: removing it leaves the single intrinsic edge (1), whose two channels (modes, primes) are the only mathematical content.

2. **Each budget is analyzed from the construction**, not guessed. The precision budget follows from elementary round-off analysis; the prime ceiling from the prolate eigenvalue asymptotic; and the mode budget from the analyticity of the Weil-form eigenvector.

3. **Every claim carries an explicit rigor level** (Table 2), separating the rigorously proved parts from those derived and validated against high-precision computation. We do not label the whole law a theorem; we state precisely which pieces are.

4. **It yields a convergence criterion** (Prop. 2): the prime cutoff is an unbeatable ceiling — at a fixed λ^2 no basis size or precision reaches the zero, so convergence to γ_1 forces $\lambda^2 \to \infty$ — which turns the qualitative convergence of [1] into an explicit, conditional rate.

5. **No prior unified law, to our knowledge.** Earlier attempts are single closed-form fits of the digit count, and they fail: we tested two proposed rate laws against high-precision data and found them unsupported [17], and Groskin [7] likewise reports that no single smooth-convergence law fits his data and declines to assert a rate, while Śliwiński [8] establishes only a lower bound on the mean error. The unified characterization given here — the intrinsic two-channel edge (1) and its finite-precision measurement (2) — is, to the best of the author's knowledge, the first working closed-form convergence law for the construction. Its unity is the point: one spectral edge (Section 7) underlies both the mode ramp and the prime ceiling, which is why a single expression can hold where earlier per-fits could not.

The deepest new content is the mode budget D_{nModes} (Section 5): we show the matching-digit ramp is the logarithm of a stretched-exponential convergence, identify its exponent with the analyticity class of the eigenvector, and derive that exponent end to end. A by-product is that the eigenvalue-space Galerkin exponent measured by Groskin [7] is a finite-window slice of the same curve. That the mode ramp and the prime ceiling are not two mechanisms but one spectral edge at two scales — the lower edge of an equilibrium measure whose bulk is the Riemann–von Mangoldt density — is the subject of Section 7.

2 The construction and the object D

We recall the construction of [1, 2]. Let $\lambda > 1$ and $L = 2\ln\lambda = \ln\lambda^2$. On $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ the functions $V_n(u) = L^{-1/2}e^{2\pi i n \ln(\lambda u)/L}$ form an orthonormal basis. The Weil quadratic form QW_λ, restricted to $\operatorname{span}\{V_n : |n| \le N\}$, is a real symmetric $(2N+1) \times (2N+1)$ matrix whose archimedean part carries the local factor at infinity and whose arithmetic part is a finite sum over prime powers $p^j \le \lambda^2$. Its smallest eigenvalue ε_N has an even eigenvector ξ, and the positive zeros of the associated Fourier–Mellin function $\widehat{\xi}$ approximate the ordinates γ_k [2].

Two facts about $\widehat{\xi}$ are used below and are due to [1, 2]: $\widehat{\xi}$ is entire, and (in the critical case) all its zeros are real and coincide with the spectrum. The eigenvector ξ is even and, in the regime we study, the smallest eigenvalue is simple — the even-symmetry verified numerically across 38 configurations spanning $\lambda^2 = 13$–1000 in [16].

The object of study, $D(\lambda^2, N, P)$, is a digit count: $D = -\log_{10}(\text{relative error})$. The relative error itself, $E = |\nu_1 - \gamma_1|/|\gamma_1|$, is the intrinsic object of the law (1); D is its measurement in digits, and the arithmetic of that measurement is what makes the minimum structure of (2) inevitable.

3 The three-budget skeleton

The computed approximation ν_1 carries error from three independent sources: finite working precision (round-off), finite basis size (mode truncation), and finite prime cutoff (the intrinsic ceiling at $N, P \to \infty$):

$$\text{error} \approx \varepsilon_{\text{round}}(P) + \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2).$$

They are independent in that any one can be made dominant by taking the other two parameters large.

Proposition 1 (The minimum is forced)**.** *For positive errors $\varepsilon_1, \varepsilon_2, \varepsilon_3$ with $\Sigma = \sum_i \varepsilon_i$,*

$$-\log_{10}\Sigma = -\log_{10}\big(\max_i \varepsilon_i\big) + O(1) = \min_i\big(-\log_{10}\varepsilon_i\big) + O(1),$$

since $\max_i \varepsilon_i \le \Sigma \le 3\max_i \varepsilon_i$ *and* $\log_{10} 3 < 0.48$*. Hence* $D = \min(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}) + O(1)$.

This is a rigorous consequence of two facts only: that D is a digit count, and that the three sources are independent. No fitting is involved. By the bound, the minimum agrees with D to within $\log_{10} 3 < 0.48$ of a digit, so the law's content is that the *binding mechanism* changes across the parameter space — round-off, mode truncation, then prime content — not that D is literally non-smooth (it is smooth: a negative logarithm of a sum).

The three sources are independent because they answer to different parameters — round-off to P, truncation to N, and prime content to λ^2 — so any one can be driven below the others at will. The $O(1)$ term is at most $\log_{10} 3 < 0.48$ of a digit, and arises only in the narrow crossover where two budgets are comparable; away from those seams a single budget binds and D equals it to within rounding. This is why, in digit space, D is a minimum rather than a sum or product, and why global single-expression fits of the digit count fail in practice: a single elementary term cannot simultaneously hold a hard precision wall, a prime ceiling, and a λ^2-coupled mode ramp. In the intrinsic form the error is, by contrast, literally a sum, $E = \varepsilon_{\text{modes}} + \varepsilon_{\text{primes}}$ (1); the digit-space minimum is what the negative logarithm of that sum becomes, the largest term binding. The junctions differ in character. The precision junction is a hard wall — round-off is a clamp — so its corner is sharp. The mode$\to$prime junction, by contrast, is a wide, smooth saturation (the tanh of Section 5): D_{nModes} is the mode-limited digit count, which rises along its pre-saturation ramp and is capped by the ceiling, $D_{\text{nModes}} \le D_{\text{Primes}}$, with equality as $N \to \infty$.

Each budget is isolated experimentally by *starvation*: to read D_i directly, set the other two parameters large enough that their budgets exceed D_i. We use this throughout Sections 4–6.

3.1 The two forms of the law

Of the three budgets, only two — mode truncation and prime cutoff — are properties of the finite operator; round-off is a property of the computation. Separating them gives the two forms stated in Section 1.

Intrinsic (spectral) form. The mathematical content is the two-channel error (1), $E = \varepsilon_{\text{modes}} + \varepsilon_{\text{primes}}$, with $\varepsilon_{\text{modes}} \sim \exp(-aN^q) \to 0$ as $N \to \infty$ (Section 5) and $\varepsilon_{\text{primes}} \to 0$ as $\lambda^2 \to \infty$ (Section 6). It makes no reference to precision, and $\nu_1 \to \gamma_1$ is exactly $E \to 0$, which requires both channels to vanish.

Observed (digit) form. A computation at working precision P resolves E only down to its round-off floor, so the measured matching-digit count is

$$D(\lambda^2, N, P) = \min\big(D_{\text{Wprec}}(P),\, -\log_{10} E(N, \lambda^2)\big) = \min\big(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}\big) + O(1), \quad (3)$$

the second equality by Proposition 1. Here $D_{\text{Wprec}}(P)$ is the instrument resolution: it is no property of ν_1, γ_1, or the operator, and as $P \to \infty$ it disappears, returning the intrinsic form in digit units. Thus the boxed law (2) is the finite-precision *measurement* of the intrinsic law (1).

One edge, not two budgets. The mode and prime channels are the two ends of a single spectral edge. Dividing out the ceiling (Section 5) writes $D_{\text{nModes}} = D_{\text{Primes}} \cdot g$ with $g = \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big) \uparrow 1$ as $N \to \infty$, so $D_{\text{Primes}} = \lim_{N\to\infty} D_{\text{nModes}}$ is the $N \to \infty$ ceiling of the mode ramp, not a separate term; equivalently the digit law is a single edge function $D_{\text{edge}}(N, \lambda^2) = D_{\text{Primes}} \cdot g$ capped by precision, $D = \min(D_{\text{Wprec}}, D_{\text{edge}}) + O(1)$. Correspondingly the intrinsic error (1) is one edge: $\varepsilon_{\text{modes}}$ is its finite-N transition (the Landau–Widom plunge, Section 5) and $\varepsilon_{\text{primes}}$ is its extreme tail (the prolate asymptotic, Section 6). That these are literally the transition and the tail of one band edge — the lower edge of the archimedean equilibrium measure — is established in Section 7.

4 The precision budget D_{Wprec}

Finite working precision P (decimal digits) introduces relative round-off $\sim 10^{-P}$, amplified by the eigenproblem's condition number κ (set by the gap of ε_N to the rest of the spectrum and independent of P). Hence

$$D_{\mathrm{Wprec}}(P) = P - \log_{10}\kappa = P + O(1), \tag{4}$$

with leading coefficient exactly 1. The $O(1)$ term is implementation dependent: the toolkit used here allocates $\lceil P \cdot \ln 10/\ln 2\rceil + 16$ working bits, the 16 guard bits contributing a small positive offset. Isolating this budget (large λ^2 and N) and sweeping $P \in \{100, 200, 300, 400, 700, 1000\}$ gives a measured slope $\partial D/\partial P = 1.000 \pm 0.001$. The exact offset is not a property of the construction and we do not assign it physical meaning. *Status: T1; the coefficient* 1 *follows from elementary numerical analysis.*

4.1 Symmetry, positivity, and the precision floor

A question the precision floor settles — and one easy to answer wrongly without it — is whether the even symmetry of the ground eigenvector, and the positivity of ε_N, are genuine features of the operator or merely of the regime above the floor. The CCM construction [1] posits a simple smallest eigenvalue with an *even* eigenvector (its Step 1) and enforces this with a forced-even projection. The reproduction of [16] tested that projection directly: across 38 configurations ($\lambda^2 = 13$–1000, $N = 10$–800, spanning both sides of the precision floor) with the projection *disabled*, the natural smallest eigenvector converged to the very vector the projection would have produced — to *bit-identical* eigenvalues — at every configuration above the floor. The forced-even projection is thus empirically a no-op above the floor: even symmetry is a property the operator exhibits unbidden, not an artifact of the projection or of the above-floor regime.

The qualifier "above the floor" is essential, and it is precisely where the budget of this section bites. When the target eigenvalue lies below D_{Wprec} — smaller than the round-off that the working precision can resolve — the computed eigenvector is dominated by numerical noise: it can present spurious mixed symmetry, and the smallest computed eigenvalue can carry a spurious sign. These are artifacts of precision starvation, not properties of the operator; raising P past the floor dissolves them, and the natural eigenvector is again even, into the deep near-zero spectrum.

This even, positive ground state was documented in our reproduction [16]. In subsequent correspondence, and in a revision of his own paper [7], Groskin independently corroborated it by a different route: the sign changes his separate quadrature implementation had shown among the deepest eigenvalues at a finite archimedean cutoff T were identified as a finite-T artifact that vanishes as T is increased, consistent with the naturally even, positive ground state of [16]. Working precision P and archimedean cutoff T are therefore independent controls: a deep-spectrum value is trustworthy only when it is stable across two settings of *each*, not when it merely holds steady in one. The even, positive ground state persists into the deep near-zero spectrum; apparent breakdowns are finite-precision or finite-cutoff artifacts, not features of the construction.

5 The mode budget D_{nModes}

This is the convergence rate in N — the open problem of [1]. In the intrinsic form it is the mode channel $\varepsilon_{\mathrm{modes}}$; as a digit count it is the rising transition of the equilibrium-measure edge (Section 7), climbing toward the prime ceiling. Dividing out that ceiling (Section 6), write the ramp ratio $g = D_{\mathrm{nModes}}/D_{\mathrm{Primes}} \in [0, 1]$. We claim

$$D_{\mathrm{nModes}}(N, \lambda^2) = D_{\mathrm{Primes}}(\lambda^2) \cdot \tanh\Big(\big(c\,N/(\lambda^2 \ln \lambda^2)\big)^{q(\lambda^2)}\Big), \qquad q(\lambda^2) = \frac{2}{2\gamma + \ln\ln\lambda^2}. \tag{5}$$

We build this in four steps: the stretched-exponential meaning of the ramp (5.1), the archimedean leading rate (5.2), the prime correction giving q (5.3), and the Shannon coordinate (5.4).

5.1 The ramp is a stretched exponential

Theorem 1 (Stretched-exponential convergence and rate unification). *Suppose the pre-saturation mode budget satisfies* $D_{\text{nModes}} = (a/\ln 10)\, N^q(1 + o(1))$ *for some* $a = a(\lambda^2) > 0$ *and exponent* $q = q(\lambda^2)$*, and that the Weil-form ground eigenvalue is simple with even eigenvector. Then:*

(i) (Stretched-exponential.) *The eigenvalue/zero error obeys* $|\nu_1 - \gamma_1|/|\gamma_1| = \exp(-aN^q)(1 + o(1))$*; equivalently* q *is the analyticity class of the eigenvector (*$q = 1$ *exponential / strip-analytic,* $0 < q < 1$ *sub-exponential but faster than any polynomial,* $q \to 0$ *finite Sobolev).*

(ii) (Rate unification.) *Any local power-law fit* $\text{err} \sim C\, N^{-2s}$ *over a window about* N *returns the* growing *local exponent* $2s_{\text{loc}}(N) = -\,d \ln \text{err}/d \ln N = a\, q\, N^q$*. Hence the eigenvalue-space Galerkin exponent* $s(c)$ *of [7] is the local logarithmic slope of this same stretched-exponential — not a fixed Sobolev index — and the digit-space ramp and the eigenvalue-space rate are one object in two coordinates. The factor* 2 *is the eigenvalue-approximation identity* $|\varepsilon - \varepsilon_N| = O(\|\xi - \xi_N\|^2)$ *for a self-adjoint operator with simple ground state [15].*

Proof. (i) $D = -\log_{10}(\text{err}) \Rightarrow \text{err} = 10^{-D} = \exp(-aN^q)$. (ii) $2s_{\text{loc}} = -\,d\ln \text{err}/d\ln N = d(aN^q)/d\ln N = aqN^q$; the squaring is the Rayleigh-quotient second-order identity at a simple eigenvalue [15]. A degenerate ground state would give only first-order dependence and no factor 2. □

Thus q is not a fitting knob but the *analyticity class* of the eigenvector: $q = 1$ is pure exponential decay (analytic in a strip), $0 < q < 1$ is sub-exponential but faster than any polynomial, and $q \to 0$ is finite Sobolev (N^{-2s}). The measured exponents lie just below 1 and decrease with λ^2 (Table 1), the signature of an eigenvector that is strip-analytic in its archimedean part and roughened by the prime content.

The force of part (i) is that it converts the open problem of *measuring* a convergence rate into the analytic question of *identifying* the eigenvector's regularity: the single exponent q fixes the entire pre-saturation curve. It also explains why fixed power-law (Sobolev) fits are structurally the wrong model — they are the $q \to 0$ limit, whereas the operator sits near $q = 1$ — and hence why such fits drift and resist extrapolation (Section 9).

Part (ii) reconciles the two published pictures: the Galerkin exponent $s(c)$ measured by Groskin [7] in eigenvalue space is the local logarithmic slope of the present stretched-exponential, sampled at his window, so it is not a fixed Sobolev index and must grow with N — his eigenvalue-space exponent and the digit-space ramp are one object in two coordinates. The squaring that links them needs the ground eigenvalue to be *simple*, which is not assumed but established as the naturally-even, isolated ground state verified across 38 configurations in the reproduction [16]; were the state degenerate, the eigenvalue error would be only first order in the eigenvector error and the factor 2 — equivalently the $2s$ in Groskin's rate — would not appear. *Status: T1 (given the pre-saturation ramp form as hypothesis).*

5.2 The archimedean leading rate $\delta = \pi/4$

The first positive zero ν_1 of the Fourier–Mellin function $\widehat{\xi}(z)$ converges to γ_1 at a rate set by the analyticity strip of the *limiting* Fourier–Mellin function, governed by the archimedean factor of the Weil form. That factor is the completed real Gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$: its logarithmic derivative $\psi(\frac{1}{4} + \frac{iz}{2}) \sim \ln(z/2)$ is precisely the $\ln|z|$ multiplier identified by Suzuki [5], and the factor itself decays as $|\Gamma(\frac{1}{4} + \frac{iz}{2})| \sim \exp(-\pi|z|/4)$ by Stirling. Hence the strip half-width is

$$\delta = \pi/4. \tag{6}$$

This value has a clean structural origin, one that also shows why the scales $\frac{1}{2}$ and $\pi/4$ both attach to the same archimedean factor. On the critical line the archimedean log-derivative has the large-z expansion

$$\psi\big(\tfrac14 + \tfrac{iz}{2}\big) = \log\frac{z}{2} + i\,\frac{\pi}{2} + o(1), \qquad z \to +\infty, \tag{7}$$

since the argument $\frac14 + \frac{iz}{2}$ runs in the imaginary (critical-line) direction, so $\arg\big(\frac14 + \frac{iz}{2}\big) \to \pi/2$. Its real and imaginary parts carry the two scales:

- the *real* part $\operatorname{Re}\psi(\frac14 + \frac{iz}{2}) \to \log(z/2)$ is the symbol, Suzuki's $\log|z|$ multiplier [5], whose analyticity strip has half-width $\frac12$ (the digamma poles at $z = \pm i/2$);
- the *imaginary* part $\operatorname{Im}\psi(\frac14 + \frac{iz}{2}) \to \frac{\pi}{2}$ sets the envelope, through

$$\frac{d}{dz}\log\big|\Gamma_{\mathbb{R}}(\tfrac12 + iz)\big| = -\tfrac12 \operatorname{Im}\psi\big(\tfrac14 + \tfrac{iz}{2}\big) \longrightarrow -\tfrac12\cdot\tfrac{\pi}{2} = -\tfrac{\pi}{4}.$$

The strip half-width and the envelope rate are therefore the real and imaginary parts of the *one* analytic function $\log\Gamma_{\mathbb{R}}(\frac12 + iz)$, a Hilbert (Kramers–Kronig) conjugate pair since $\Gamma_{\mathbb{R}}$ is zero-free (minimum phase): its log-magnitude and phase determine each other. In one identity,

$$\delta = \frac{\pi}{4} = \frac12\cdot\frac{\pi}{2} = \tfrac12\arg(i), \tag{8}$$

half the right angle of the critical-line direction. The two scales $\frac12$ and $\pi/4$ are thus not independent constants but conjugate parts of the single factor $\Gamma_{\mathbb{R}}$: the strip half-width $\frac12$ and its harmonic conjugate $\pi/4$.

Two points delimit what this establishes. It fixes the *value* $\delta = \pi/4$ as a property of the limiting $\Gamma_{\mathbb{R}}$ factor: δ is the half-width of the strip carrying the zeros of the limiting Fourier–Mellin function, and governs the zero convergence $\nu_1 \to \gamma_1$ (with the squaring of Thm. 1(ii)); it is not the decay rate of the finite eigenvector's Galerkin coefficients, a distinct band-limited, prime-modulated quantity (≈ 0.71 at $\lambda^2 = 100$), and the finite $\widehat{\xi}$, of exponential type $L/2$ via its $\sin(zL/2)$ factor, is not literally the completed ξ. And it is structural motivation rather than a full derivation: obtaining $\pi/4$ from the symbol alone would require its conjugate part, which appears only through the analytic completion $\Gamma_{\mathbb{R}}$, that is, through the finite ground-state transform inheriting that factor, the realization step that remains open. *Status: T3.*

Lemma 1 (Lattice sampling of the coefficients)**.** *The Fourier–Mellin transform of the N-mode eigenvector has the closed form $\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)\sum_{|j|\le N}\xi_j/(z - 2\pi j/L)$ [2]. On the frequency lattice $z_k = 2\pi k/L$,*

$$\widehat{\xi}_N(2\pi k/L) = (-1)^k\sqrt{L}\,\xi_k \quad (|k|\le N), \qquad \widehat{\xi}_N(2\pi k/L) = 0 \quad (|k| > N),$$

so $|\xi_k| = L^{-1/2}\big|\widehat{\xi}_N(2\pi k/L)\big|$, with $L = \ln\lambda^2$ the period of the basis $\{V_n\}$.

Proof. At $z_k = 2\pi k/L$ the factor $\sin(zL/2) = (-1)^k\sin\big((z - z_k)L/2\big)$ has a simple zero, so every $j \ne k$ term vanishes and the $j = k$ pole is removable: $\lim_{z\to z_k} 2L^{-1/2}(-1)^k\sin\big((z-z_k)L/2\big)/(z - z_k) = (-1)^k\sqrt{L}$. For $|k| > N$ there is no $j = k$ term and the common factor $\sin(z_kL/2) = 0$. □

By Lemma 1 the coefficients are the lattice samples of $\widehat{\xi}_N$ at spacing $2\pi/L$ with $L = \ln\lambda^2$, so the strip half-width (6) gives a coefficient tail $\exp(-\delta(2\pi/L)N)$; with the squaring of Thm. 1(ii) the leading digit slope is

$$\frac{\partial D_{\text{nModes}}}{\partial N} = \frac{2\delta\,(2\pi/L)}{\ln 10} = \frac{\pi^2}{\ln 10\ \ln\lambda^2} \approx \frac{4.286}{\ln\lambda^2}, \tag{9}$$

a parameter-free prediction. At $\lambda^2 = 7$, where few primes enter and $q \approx 1$, (9) gives 2.203 against a measured small-N slope of 2.264 — agreement to 3% with no free parameters. At larger λ^2 the measured early slope runs above (9) exactly as the $q < 1$ concavity (below) requires.

Table 1: Per-λ^2 fits of (5) (full tanh form, mode-limited points). The onset constant c clusters near $4/\pi = 1.273$; the exponent q decreases through 1 near $\lambda^2 \approx 13$, matching the archimedean crossover. $\lambda^2 = 200$ (sparse, 8 points) is shown but excluded from the regression.

λ^2	7	13	20	25	50	100	150	200
q	1.185	0.985	0.900	0.880	0.835	0.785	0.760	0.805
c	1.20	1.25	1.28	1.29	1.31	1.28	1.26	1.31

5.3 The prime correction: $q(\lambda^2)$

The leading rate gives $q = 1$. The finite Euler product $\prod_{p\leq\lambda^2}(1 - p^{-s})^{-1}$ modulates the archimedean envelope by an oscillatory prime term $S(z) = \sum_{p\leq\lambda^2} p^{-1/2}\cos(z\ln p) + \cdots$ of mean-square strength

$$\langle S^2\rangle = \tfrac{1}{2}\sum_{p\leq\lambda^2} p^{-1} = \tfrac{1}{2}(\ln\ln\lambda^2 + M) + o(1)$$

by Mertens' theorem (M the Mertens constant); equivalently the full log-product is $\sum_{p\leq\lambda^2} -\ln(1 - 1/p) = \gamma + \ln\ln\lambda^2$. More primes roughen the eigenvector and lower the analyticity exponent below 1. Modeling the reciprocal exponent as a baseline plus the prime fluctuation,

$$\frac{1}{q} = c_0 + c_1\ln\ln\lambda^2, \qquad c_0 = \gamma \text{ (archimedean baseline)}, \quad c_1 = \tfrac{1}{2} \text{ (variance prefactor)}, \quad (10)$$

which gives $q(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$. The numerator 2 is $1/c_1$, the reciprocal of the $\frac{1}{2}$ in the mean square; the $\ln\ln$ is Mertens.

We test (10) against per-λ^2 fits of the full form (5) (Table 1). A linear regression $1/q = c_0 + c_1\ln\ln\lambda^2$ over $\lambda^2 \in \{7, 13, 20, 25, 50, 100, 150\}$ yields

$$c_0 = 0.556\ (\gamma = 0.577), \qquad c_1 = 0.478\text{–}0.508\ (\tfrac{1}{2} = 0.500), \qquad R^2 = 0.98.$$

Euler–Mascheroni γ appears as the intercept of a regression that does not assume it. *Status: T3; the remaining lemma is a rigorous variance-to-exponent map.*

5.4 The Shannon coordinate

Lemma 2 (Coordinate from ceiling over slope)**.** *The natural coordinate of the ramp is $N/(\lambda^2\ln\lambda^2)$. The scale at which the leading ramp meets its ceiling is*

$$N^* = \frac{D_{\text{Primes}}}{\partial D/\partial N} = \frac{(4\pi/\ln 10)\lambda^2}{\pi^2/(\ln 10\ \ln\lambda^2)} = \frac{4}{\pi}\lambda^2\ln\lambda^2.$$

The λ^2 comes from the prime ceiling (Section 6); the $\ln\lambda^2$ from the inverse of the archimedean slope (9). The combination $\lambda^2\ln\lambda^2$ — the time–frequency Shannon number of the box (Landau–Pollak [12]; Slepian–Pollak [11]; Landau–Widom [10]) — is therefore forced, not posited. (Measured saturation occurs at $N/(\lambda^2\ln\lambda^2) \approx 2.0$–$2.3$, larger than the linear value $4/\pi$ because the $q < 1$ ramp reaches its ceiling later than the initial-slope line.) The form and the constant separate cleanly: writing $N^* = \lambda^2\ln\lambda^2/\delta$, the grouping $\lambda^2\ln\lambda^2$ is forced by the ceiling's λ^2 scaling and the basis interval $L = \ln\lambda^2$ (Lemma 1) alone — independently of the archimedean constant — while only the multiplier $4/\pi = 1/\delta$ carries the rate δ. That multiplier is the data-pinned piece: the full-ramp onset constant clusters at $4/\pi$ across the cutoffs of Table 1 (mean ≈ 1.27). *Status: the coordinate form is T2 (ceiling-over-slope inputs); the multiplier $4/\pi$ is T3, data-pinned.*

6 The prime ceiling D_{Primes}

The prime channel of the intrinsic law (1) is the accuracy at infinite N and P, set solely by which primes $p \leq \lambda^2$ enter the form. It is the relative error of the first zero at the cutoff,

$$\varepsilon_{\mathrm{primes}}(\lambda^2) = \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{11}$$

with $\nu_1(\infty, \lambda^2)$ the first zero at infinite basis size and fixed λ^2: it is what remains once modes and precision are exhausted, and its digit measurement $D_{\mathrm{Primes}} = -\log_{10} \varepsilon_{\mathrm{primes}}$ is the ceiling the minimum (2) consumes. We fix its leading rate through the plunge *eigenvalue*, the tractable route, kept here in digit space to reinforce the intrinsic statement with the measured quantity.

The plunge eigenvalue gives an *eigenvalue floor* $D_{\mathrm{eig}} = -\log_{10} \varepsilon^{\infty}(\lambda)$, the digit value of the smallest eigenvalue. The CCM construction ties ε^{∞} to the principal prolate spheroidal eigenvalue at bandwidth λ; the prolate eigenvalue asymptotic (Connes [4], §6.4; Fuchs [9]; cf. Connes [3] and Hedenmalm [6]) gives, for the relevant index,

$$1 - \chi(\lambda) \sim \frac{2^{14}\sqrt{2}\,\pi^5}{3} \exp\!\big(-4\pi\lambda^2 + \tfrac{9}{2}\ln\lambda^2\big),$$

so that

$$D_{\mathrm{eig}}(\lambda^2) = K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0, \qquad K = \frac{4\pi}{\ln 10} = 5.45750\ldots, \tag{12}$$

with every constant closed-form: the leading $K = 4\pi/\ln 10$ is the base-10 rendering of the $e^{-4\pi\lambda^2}$ prolate decay rate that appears in Connes' own framework, the $\frac{9}{2}$ subleading-log coefficient is the Fuchs index, and $C_0 = \log_{10}\!\big(2^{14}\sqrt{2}\,\pi^5/3\big) = 6.3736\ldots$ follows from the prolate asymptotic (Groskin [7] independently evaluates the same §6.4 prefactor and obtains $\log_{10} \approx 6.37$, in agreement). This fixes the leading rate of $\varepsilon_{\mathrm{primes}}$: the plunge sets $K\lambda^2$.

The eigenvalue depth is a scaffold, not the object. What the law consumes is the zero error $\varepsilon_{\mathrm{primes}}$, which sits a few digits below the eigenvalue floor: converting the plunge depth ε^{∞} into the zero displacement $|\nu_1 - \gamma_1|$, and normalizing by $|\gamma_1|$ (the $\log_{10}|\gamma_1| \approx 1.15$ digits of the relative error), gives the matching floor

$$D_{\mathrm{Primes}}(\lambda^2) = K\lambda^2 - P_m \log_{10}\lambda^2 - c_m, \qquad P_m \approx 5, \quad c_m \approx 9.5. \tag{13}$$

The leading $K\lambda^2$ is shared with the eigenvalue floor, the same plunge rate, so the intrinsic content is a *single* channel, not two competing ceilings; the subleading P_m, c_m are the eigenvalue-to-zero conversion together with the $|\gamma_1|$ normalization. Equivalently the *conversion gap* $\mathrm{conv}(\lambda^2) := D_{\mathrm{eig}} - D_{\mathrm{Primes}} = (P_m - \frac{9}{2})\log_{10}\lambda^2 + (c_m - C_0) \approx \frac{1}{2}\log_{10}\lambda^2 + 3$ is that conversion, read in digit space. The leading rate K is derived (prolate transfer); the conversion, hence P_m and c_m, is empirical — of it, the ≈ 1.15-digit part of c_m is the explicit $|\gamma_1|$ normalization and the remainder is the depth-to-zero factor.

The arithmetic does not move the eigenvalue-floor slope. At the scale the operator resolves, the prime part of the Weil form has large-$|t|$ behavior $S(t) = \frac{1}{2}t^2 - \gamma|t| + c_0$ (Mertens' theorems with prime-number-theorem partial summation). On the even, mean-zero ground eigenvector each piece is inert on the prefactor power: $\frac{1}{2}t^2$ acts as a rank-one *odd* operator that annihilates the even ground state; $-\gamma|t|$ equals $2\gamma K_a$, a multiple of the metric operator $K_a = (-\Delta)^{-1}$, so it only rescales the eigenvalue and shifts the additive constant; and c_0 projects out of the mean-zero subspace (the screw-function operator split of [5]). None of the leading prime terms can therefore supply a $(\lambda^2)^{\Delta}$ prefactor correction: given the rate, the prefactor power is the Fuchs index $\frac{9}{2}$ of the prolate model (Section 6), and the entire arithmetic contribution is confined to the additive constant. This inertness is a statement about the prefactor only — the archimedean part of the form does not by itself carry the plunge (it is $O(1)$-indefinite; see the rate-origin discussion in Section 9). The empirically steeper matching-floor slope $P_m \approx 5$ is a finite-λ^2 curvature effect, not a change in the asymptotic slope.

Status: the leading coefficient K and the eigenvalue-floor slope $\frac{9}{2}$ ride the prolate transfer (T2), and the arithmetic is provably inert on the slope; only the matching-floor offset (P_m, c_m), the eigenvalue-to-zero conversion, is empirical (T4).

The floor is smooth, not a prime staircase. A natural hypothesis — that D_{Primes} steps at prime entries — is falsified directly. With N and P held above floor ($N = 300$, $P = 200$), sweeping λ^2 in unit steps from 4 to 30 gives a smooth rise of ≈ 5.3 digits per unit λ^2; the per-unit rate at prime-power entries is statistically indistinguishable from non-prime entries. The ceiling is a smooth function of λ^2, consistent with the analytic form (13).

Floor anchors. Floor-isolated measurements (both N and P above floor) give $D_{\text{Primes}}(100) = 526.08$ at $(N, P) = (1000, 700)$ and a saturated $D_{\text{Primes}}(200) \approx 1070.4$ at $N \geq 2400$. The latter sits at the matching floor $K \cdot 200 - 5 \log_{10} 200 - 9.5 = 1070.5$, confirming the two-floor structure at the largest cutoff measured.

7 The equilibrium-measure edge

Sections 5 and 6 derived the mode ramp and the prime ceiling by two separate classical routes: the Landau–Widom transition width for the ramp, and the Fuchs prolate tail for the ceiling. This section states the structural fact that ties them together — both are features of a single spectral edge — so the intrinsic law (1) is one edge rather than two coincident mechanisms. It adds no new constant; it is the reason the two budgets share a form.

7.1 The archimedean equilibrium measure

Treat the limiting ordinates $\{\nu_k\}$ as a unit charge in the external field set by the archimedean weight of the Weil form. As $\lambda^2 \to \infty$ the macroscopic distribution is the equilibrium measure μ_Q minimizing the weighted logarithmic energy

$$I[\mu] = \iint \log \frac{1}{|s-t|}\, d\mu(s)\, d\mu(t) + 2 \int Q(t)\, d\mu(t), \qquad \mu \geq 0,\ \textstyle\int d\mu = 1,$$

with external field Q the archimedean potential whose derivative is the phase rate $Q'(t) = \frac{1}{2}\log(t/2\pi)$ of Section 5.2. That a limiting spectral or zero density is the equilibrium measure of such a weighted energy problem is classical (Saff–Totik [13]; in the orthogonal-polynomial and random-matrix setting, Deift [14]); we invoke it here as a known tool. The Euler–Lagrange condition $2\int \log|s-t|\, d\mu(t) = 2Q(s) + \text{const}$ on $\operatorname{supp}\mu$ fixes the density; with this field it is

$$\rho_Q(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}, \tag{14}$$

the Riemann–von Mangoldt zero-counting density ($\int_0^T \rho_Q = \frac{T}{2\pi} \log \frac{T}{2\pi e}$). The smooth backbone of the spectrum is thus the equilibrium measure of the archimedean field, recovered without reference to the primes and independent of RH. *Status: T3 — the density* (14) *is the standard archimedean main term and the variational identification is structural; a rigorous equilibrium-measure limit for the truncated operator is the open step.*

7.2 One edge, two scales

The equilibrium measure has a lower band edge: the eigenvalues descend from the $O(1)$ bulk into the deep, positive plunge (positivity forced by the Weil form). That edge is a single object, and the two budgets are its two scales:

- *the transition is the mode ramp* — the width of the crossover from bulk to plunge, as the resolved dimension crosses the Shannon number $\lambda^2 \ln \lambda^2$, is the Landau–Widom log-width [10], which is the tanh ramp of D_{nModes} (Section 5);

- *the extreme tail is the prime ceiling* — the depth past the edge is the Fuchs extreme-value asymptotic [9], which is D_{Primes} (Section 6).

So the Landau–Widom width and the Fuchs tail, invoked separately above, are the transition and the tail of the *same* band edge. This is the mechanism behind the algebra $D_{\mathrm{nModes}} = D_{\mathrm{Primes}} \cdot g$ of Section 3.1: D_{nModes} and D_{Primes} are one edge D_{edge} read at two resolutions, and the intrinsic error (1) is correspondingly one edge — its finite-N transition $\varepsilon_{\mathrm{modes}}$ and its extreme tail $\varepsilon_{\mathrm{primes}}$ are the ramp and the ceiling of μ_Q's lower edge. The unification is conceptual, not a new estimate: the microscopic universality class of the edge (Airy, Bessel, or a Landau–Widom band edge) and the exact deep-edge constant remain open, the latter being the same archimedean–prime cancellation left open for D_{Primes} in Section 6. The equilibrium-measure framework itself is classical (above); to the best of the author's knowledge, its application to the Connes–Consani–Moscovici / Connes–van Suijlekom Weil form — identifying the zero-counting backbone as the archimedean equilibrium measure, and the mode ramp and prime ceiling as the transition and extreme tail of that measure's lower edge — is new here. *Status: T3 — structural unification riding the prolate transfer already used for both budgets; the framework is classical and no new constant is claimed.*

8 The complete formula

Collecting Sections 4–7, the law stands in the two forms of the introduction. *Intrinsically* it is the single edge of Section 7: the relative error is the two-channel quantity

$$E(N, \lambda^2) = \varepsilon_{\mathrm{modes}}(N, \lambda^2) + \varepsilon_{\mathrm{primes}}(\lambda^2), \tag{15}$$

equivalently, in digit units, the edge function $D_{\mathrm{edge}}(N, \lambda^2) = D_{\mathrm{Primes}}(\lambda^2) \cdot \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big)$, whose $N \to \infty$ ceiling $D_{\mathrm{edge}}(\infty, \lambda^2) = D_{\mathrm{Primes}}$ is the prime floor and whose finite-N transition is the mode ramp D_{nModes}. *Observed* at working precision P, it is that one edge capped by the instrument resolution:

$$D(\lambda^2, N, P) = \min\Big(\underbrace{P + O(1)}_{D_{\mathrm{Wprec}}},\ \underbrace{D_{\mathrm{Primes}} \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big)}_{D_{\mathrm{nModes}} = D_{\mathrm{edge}}},\ \underbrace{K\lambda^2 - P_m \log_{10} \lambda^2 - c_m}_{D_{\mathrm{Primes}} = D_{\mathrm{edge}}(\infty)}\Big), \tag{16}$$

with $K = 4\pi/\ln 10$ exact, $q = 2/(2\gamma + \ln \ln \lambda^2)$, and $c \approx 4/\pi$. The last two entries are the one edge D_{edge} at finite N and at $N \to \infty$, so the operative law is $D = \min(D_{\mathrm{Wprec}}, D_{\mathrm{edge}}) + O(1)$: a single spectral edge, capped by precision. This is the Andrews CCM Convergence Law. Of its parts, the minimum structure, the precision slope, and the stretched-exponential convergence and rate unification (Thm. 1) are rigorous (T1); the Shannon coordinate and the prime leading coefficient are derived modulo a stated lemma (T2); the archimedean rate, the exponent q, and the onset constant are derived in structure and validated to a few percent against high-precision data (T3); the prime subleading offset is empirical (T4). Table 2 is the precise ledger.

To the best of the author's knowledge, the complete convergence model of (16) is novel, meaningful, and has not been previously published: the assembly of the three budgets into a single law, the closed-form mode exponent $q(\lambda^2) = 2/(2\gamma + \ln \ln \lambda^2)$, the Shannon coordinate $\lambda^2 \ln \lambda^2$, the stretched-exponential identification of the mode budget, and the recognition of the mode ramp and prime ceiling as one equilibrium-measure edge appear here for the first time. The rate it characterizes is precisely the problem named open in [1]: Groskin [7] measures a Galerkin exponent but disclaims deriving it, and Śliwiński [8] establishes only a lower bound, so no prior work supplies the functional form given here. The novelty claimed is the law as a whole and the identification of its mechanism; the underlying construction [1, 2], the Paley–Wiener strip mechanism [7], and the classical prolate and Mertens inputs are credited to their sources throughout.

Proposition 2 (Convergence criterion and the prime bottleneck)**.** *Let $\nu_1 = \nu_1(\lambda^2, N, P)$ and $D = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$.*

(i) (Prime bottleneck.) *For every fixed λ^2, $\sup_{N,P} D(\lambda^2, N, P) = D_{\text{Primes}}(\lambda^2) < \infty$. Hence no sequence with bounded λ^2 has $\nu_1 \to \gamma_1$: convergence to the zero forces $\lambda^2 \to \infty$.*

(ii) (Criterion.) *Since $D_{\text{nModes}} \leq D_{\text{Primes}}$ — a mode-limited count cannot beat the ceiling — $D \to \infty$ if and only if $D_{\text{Wprec}} \to \infty$ and $D_{\text{nModes}} \to \infty$; the latter forces $\lambda^2 \to \infty$ (part (i)), with N past the Shannon scale $\lambda^2 \ln \lambda^2$ and $P \to \infty$. The mode budget rises monotonically toward its ceiling as $N \to \infty$ ([1], Prop. 3.4).*

Consequently, conditional on the continuum plunge $D_{\text{Primes}}(\lambda^2) \to \infty$ (i.e. $\varepsilon^\infty(\lambda) \to 0$), convergence holds along any path $\lambda^2 \to \infty$ with $N \gtrsim (4/\pi)\lambda^2 \ln \lambda^2$ and $P \gtrsim K\lambda^2$, at rate $D \sim K\lambda^2$.

Proof. (i) As $N \to \infty$ the truncation converges to the continuum operator at cutoff λ, whose first zero lies at relative distance $10^{-D_{\text{Primes}}(\lambda^2)}$ from γ_1; here $D_{\text{Primes}}(\lambda^2) < \infty$ because $\varepsilon^\infty(\lambda) > 0$ (Weil positivity; [8]) and is N-independent ([5], Thm 1.3), and finite working precision only lowers D. Hence $D \leq D_{\text{Primes}}(\lambda^2)$, finite, for all N, P. (ii) By Prop. 1, $D = \min(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}) + O(1)$; since $D_{\text{nModes}} \leq D_{\text{Primes}}$, this is $\min(D_{\text{Wprec}}, D_{\text{nModes}}) + O(1)$, so $D \to \infty$ if and only if $D_{\text{Wprec}} \to \infty$ and $D_{\text{nModes}} \to \infty$; since $D = -\log_{10}(|\nu_1 - \gamma_1|/|\gamma_1|)$, this is exactly $\nu_1 \to \gamma_1$. □

This refines the qualitative monotone convergence of [1] (Prop. 3.4) into an explicit, conditional rate. The equivalence $D \to \infty \Leftrightarrow \nu_1 \to \gamma_1$ and the prime bottleneck (i) are rigorous (T1); the conditional rate inherits the rigor of D_{Primes} — the prolate transfer (T2) and the continuum plunge it assumes.

Conceptually, the criterion isolates the prime cutoff as the sole hard bottleneck: precision and basis size are freely improvable and impose no ceiling, so reaching the exact zero is, at root, a statement about $\lambda^2 \to \infty$ and the growth of the prime content. The rate $D \sim K\lambda^2$ quantifies the cost — each additional digit demands a fixed increment $\ln 10/(4\pi)$ in λ^2 — turning the construction's qualitative convergence into a concrete, if conditional, budget.

9 Discussion

Relation to Groskin [7]. Groskin independently implemented the Connes–van Suijlekom matrix and measured a Galerkin convergence exponent $s(c)$ growing with the cutoff, proposing a Paley–Wiener analyticity-strip mechanism. The present work builds on that mechanism: Thm. 1(ii) shows his $s(c)$ is the local slope of our stretched-exponential, and the strip width is identified with the archimedean $\Gamma_{\mathbb{R}}$ rate $\delta = \pi/4$. What is new here is the digit-space formulation, the Shannon coordinate, the closed form for $q(\lambda^2)$, and the embedding of the rate in the full three-budget minimum.

Groskin's revised analysis, extended to cutoff $\lambda^2 = 100$, is independent corroboration of this two-regime structure. He finds that no single smooth-convergence law fits his eigenvalue-space data and explicitly declines to assert a convergence rate, reporting instead an unresolved tension between a *drifting* effective exponent and an apparent *finite* convergence limit. Both halves are accounted for here: Theorem 1(ii) explains the drift — a fixed Sobolev index cannot persist, since the local exponent $2s_{\text{loc}}(N) = aqN^q$ grows with N — while the prime ceiling of Section 6 supplies the finite limit $D_{\text{Primes}}(\lambda^2)$. What looks like an irreconcilable choice within a single-rate framework is simply the pre-saturation ramp and the ceiling of the minimum law (2), seen in one dataset.

Relation to Śliwiński [8] and Suzuki [5]. Śliwiński proves an inverse-logarithmic lower bound on the *mean* spectral error over the whole spectrum (Thm. 3.1) — a different regime from the first-zero convergence here, and consistent with it, since the mean is dominated by the unconverged high-index eigenvalues while $\nu_1 \to \gamma_1$ converges super-exponentially. Suzuki's screw-function analysis supplies the archimedean $\ln|z|$ multiplier and the prime-frequency support that underlie Sections 5.2–5.3.

Origin of the leading rate: arithmetic, not archimedean. The prime ceiling is *modeled* by an archimedean object — the prolate eigenvalue of Section 6 — yet the exponential rate is not carried by the archimedean part of the Weil form itself. Diagonalizing the screw-function split [5] at high precision, the archimedean-only localized form is $O(1)$-indefinite, with no exponentially small eigenvalue. Restricting it to the Sonin subspace — the orthogonal complement of the band-concentrated prolate modes, where Connes establishes archimedean Weil positivity [4] — restores positivity, but only as an $O(1)$ gap: the smallest Sonin eigenvalue is 2.26, 2.28, 2.57 at $\lambda^2 = 5, 8, 13$, essentially flat in λ^2, against a floor $K\lambda^2 = 27, 44, 71$. Archimedean positivity and the plunge are therefore *decoupled*: the place at infinity supplies an $O(1)$ spectral gap, while the exponentially small $\varepsilon^\infty \sim e^{-4\pi\lambda^2}$ is produced by the cancellation between the archimedean and prime parts of the full form. This locates the single open analytic step behind the leading coefficient: a quantitative archimedean–prime cancellation estimate on the even ground eigenvector — the precise sense in which K is T2.

Practical use. The law is a parameter-selection guide: to target D^* digits, pick $\lambda^2 \gtrsim D^* \ln 10/(4\pi) = D^*/5.46$ so the ceiling clears D^*; pick N so the tanh ramp clears D^* (a few times the Shannon number $\lambda^2 \ln \lambda^2$); and pick $P \gtrsim D^*$. Increasing any one parameter past the binding minimum yields nothing.

10 Rigor ledger

Table 2 collects the rigor status of every component of the law, together with the open item that would promote each to the next level.

Table 2: Rigor status of each component. T1: rigorous. T2: derived modulo one named lemma. T3: derived in structure and validated against data, with a map not yet rigorous. T4: empirical.

Component	Status	Open item
Minimum structure (Prop. 1)	**T1**	—
$D_{\mathrm{Wprec}} = P + O(1)$, slope 1	**T1**	implementation offset
Stretched-exp convergence + rate unification (Thm. 1)	**T1**	—
Convergence criterion (Prop. 2)	**T1**	continuum plunge $\varepsilon^\infty \to 0$ (T2)
D_{Primes} leading $K = 4\pi/\ln 10$	**T2**	prolate asymptotic transfer
D_{Primes} floor slope 9/2	**T2**	prolate transfer; primes provably inert
D_{Primes} matching offset (P_m, c_m)	**T4**	empirically pinned
Shannon coordinate $\lambda^2 \ln \lambda^2$	**T2**	ceiling/slope (Lem. 2)
Equilibrium-measure edge (one edge, two scales)	**T3**	equilibrium limit; edge universality class
Archimedean rate $\delta = \pi/4$	**T3**	zero-convergence from $\Gamma_{\mathbb{R}}$
Exponent $q = 2/(2\gamma + \ln\ln\lambda^2)$	**T3**	variance$\to$exponent map
Onset constant c ($\sim 4/\pi$)	**T3**	exact rational

11 Conclusion

We have given a single law (2) for the convergence of the CCM zeta spectral triple, in which the minimum structure is forced, the prime ceiling has the exact leading coefficient $4\pi/\ln 10$, and the mode budget — the convergence rate named open in [1] — is the logarithm of a stretched-exponential whose exponent is the eigenvector's analyticity class, derived end to end as $q = 2/(2\gamma + \ln\ln\lambda^2)$. The mode ramp and the prime ceiling are not two mechanisms but one object — the lower edge of the archimedean equilibrium measure, whose bulk density is the Riemann–von Mangoldt law — seen at two scales (Section 7). Read as a convergence statement, the law says $\nu_1 \to \gamma_1$ exactly when all three budgets diverge, giving an explicit rate $D \sim (4\pi/\ln 10)\lambda^2$ along $\lambda^2 \to \infty$ (Prop. 2). Each claim is stated with an explicit rigor level. The rigorous core (the minimum, the precision slope, and the stretched-exponential convergence theorem 1, which unifies the digit- and eigenvalue-space rates) stands on its own; the remaining lemmas — a coefficient-decay estimate for the $\Gamma_{\mathbb{R}}$-weighted form, and a variance-to-exponent map — would promote the archimedean rate and the exponent to theorems, and are the natural next steps.

12 Explicit non-claims

The per-component rigor of the law is the subject of Table 2 and is not repeated here. Beyond it, we are explicit about three boundaries.

- **No proof of the Riemann Hypothesis.** We characterize how the CCM/CvS approximation converges; the link from the construction to RH — continuum positivity of QW_λ for all λ — is the open conjecture of [1, 2, 4], and is neither assumed nor advanced here.
- **We do not establish that the construction reproduces the zeta zeros.** That the finite operator's zeros converge to the true ordinates as $\lambda^2 \to \infty$ is the construction's own open premise; we quantify the approach to γ_1 *under* that premise (Prop. 2) and do not prove the premise itself.
- **Scope.** The law is stated and tested for the first zero γ_1 of the Riemann zeta function. We make no claim about its constants for higher zeros, nor about extension to Dirichlet L-functions or other settings.

Statement on the use of AI tools

In keeping with the arXiv policy on authors' use of generative AI language tools, the author discloses that Anthropic's Claude AI model served as a text-to-text aid in preparing the LaTeX exposition and the Rust implementation code, to specifications set by the author. AI assistance was also used in gathering research material; all such material was inspected and characterized by the author. The author originated, directed, and checked every computational experiment and mathematical claim presented here, and bears full responsibility for the contents of this manuscript.

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Empirical testing and code availability

The results above rest on an extensive empirical program. The author conducted over 5,000 high-precision CCM experiments to observe the operator's behavior across the cutoff, basis-size, and working-precision axes. This testing was critical: it provided the foundation for the derivation, ruling candidate ideas in or out quickly and aiming the analysis at the behavior the operator actually exhibits.

The paper source and reproduction scripts are available at `https://github.com/TeamXcelerator/ccm-convergence-rate`; the shared high-precision library (Xcelerator Toolkit) is at `https://github.com/TeamXcelerator/xcelerator-toolkit`. Both are source-available in accordance with the licensing provided with each repository.

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